

Lab 13 Controlled Experiments:

Lab Assignment 2:

Objective:

The objective of this lab is to give hands of experience to the students on planning, executing and then analyzing the results of a controlled experiment.

Activity Outcomes:

The activities provide hands - on practice with the following topics

- Designing a controlled experiment
- Executing an experiment.
- Performing a t-test on the results.

1) Useful Concepts

Executing Controlled Experiments: You start by stating a clear, testable hypothesis. By testable, we mean that the hypothesis must be quantifiable and measurable. Here's an example of a hypothesis that we might want to test that the Macintosh menu bar, which is anchored to the top of the screen, is faster to access than the Windows menu bar, which is separated from the top of the screen by a window title bar.

You then choose your independent variables - the variables you're going to manipulate in order to test the hypothesis. In our example, the independent variable is the kind of interface: Mac menubar or Windows menubar. In fact, we can make it very specific: the independent variable is the y-position of the menubar (either $y = 0$ or $y = 16$, or whatever the height of the title bar is). Other independent variables may also be useful. For example, you may want to test your hypothesis on different user classes (novices and experts, or Windows users and Mac users). You may also want to test it on certain kinds of tasks. For example, in one kind of task, the menus might have an alphabetized list of names; in another, they might have functionally-grouped commands.

You also have to choose the dependent variables, the variables you'll actually measure in the experiment to test the hypothesis. Typical dependent variables in user testing are time, error rate, event count (for events other than errors - e.g., how many times the user used a particular command), and subjective satisfaction (usually measured by numerical ratings on a questionnaire).

Finally, you use statistical techniques to analyze how your changes in the independent variables affected the dependent variables, and whether those effects are significant (indicating a genuine cause-and-effect) or not (merely the result of chance or noise).

t-value		
	$\frac{ \bar{X}_1 - \bar{X}_2 }{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	

H₀

Degrees of Freedom	p=0.05	p=0.025	p=0.01
1	12.71	25.45	63.66
2	4.30	6.20	9.92
3	3.18	4.17	5.84
4	2.78	3.50	4.60
5	2.57	3.16	4.03
6	2.45	2.97	3.71
7	2.36	2.84	3.50
8	2.31	2.75	3.36
9	2.26	2.68	3.25
10	2.23	2.63	3.17
11	2.20	2.59	3.11
12	2.18	2.56	3.05
13	2.16	2.53	3.01
14	2.14	2.51	2.98
15	2.13	2.49	2.95
16	2.12	2.47	2.92
17	2.11	2.46	2.90
18	2.10	2.44	2.88
19	2.09	2.43	2.86
20	2.09	2.42	2.84
21	2.08	2.41	2.83
22	2.07	2.41	2.82
23	2.07	2.40	2.81
24	2.06	2.39	2.80
25	2.06	2.38	2.79
26	2.06	2.38	2.78
27	2.05	2.37	2.77
28	2.05	2.37	2.76
29	2.04	2.36	2.76
30	2.04	2.36	2.75
40	2.02	2.33	2.70
60	2.00	2.30	2.66
120	1.98	2.27	2.62

Solved Lab Activities:

Activity 1

Designing a Controlled Experiment

We want to compare the typing speed of kindle and ipad keyboard

Hypothesis: “University students (population) type (task) faster (measurement) using iPad’s keyboard (feature) than using Kindle’s keyboard”

One tailed hypothesis

Null Hypothesis: “There is no difference in speed when university students type using kindle or ipad”

IV: Type of keyboard

DV: time

Experimental Design: We would take 20 university students from different departments and then randomly assign them to one of the experimental conditions

Condition A: Type the given paragraph using ipad

Condition B: Type another paragraph of the same length using kindle.

Activity 2

Applying t-test

The following data was gathered after executing the above experiment

Subj	Kindle Time (s)	iPad Time (s)
1	43	34
2		33
3	43	36
4	35	31
5	36	41
6	39	39
7	42	5
8	43	29
9	41	30
10	39	41

Now we will apply t-test on this data

First we calculate the t-value as given in the formula above.

	Kindle	ipad
	43	34
	37	33
	43	36
	35	31
	36	41
	39	39
	42	37
	43	29
	41	30
	39	41
Mean	39.8	35.1
Standard deviation	3.047768	4.408325
Variance	9.288889	19.433333
Count	10	10

$$X1-x2 = 1.360558$$

$$\text{variance/count} = 0.928889$$

$$\text{Adding} = 2.872222$$

$$\text{Sqrt} = 1.694763$$

$$\text{t-value} = 0.802801$$

$$\text{df} = n1+n2-2 \\ = 18$$

critical
value is
1.73

Our t-value is 0.8 which is less then critical value so we cannot reject null hypothesis

You can do all these calculations in excel

Graded Lab Tasks:

Lab Task 1

Design a controlled experiment to test whether its faster to upload assignment on CUOnline or Teams. I have created dummy assignments on both platforms for you so you can test it. You have to submit individually but must collaborate with your entire class to collect data.

Write the hypothesis, IV, DV, is it one tailed or two tailed. Also design the protocol and use t-test to accept or reject your hypothesis.

You can use the following table

	one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
	two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df												
1		0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2		0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3		0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4		0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5		0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6		0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7		0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8		0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9		0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10		0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11		0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12		0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13		0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14		0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15		0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16		0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17		0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18		0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19		0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20		0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21		0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22		0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23		0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24		0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25		0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26		0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27		0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690